

$d=4$: Four external fields:

leading order contributions for

leading order contributions for

$$\langle \phi_1 \dots \phi_n \rangle \sim$$

Three Feynman diagrams representing leading-order contributions. The first diagram is a tree-level exchange with external lines 1, 2, 3, 4 and internal lines 1, 2, 3, 4. The second diagram is a loop diagram with external lines 1, 2, 3, 4 and internal lines 1, 2, 3, 4, with a blue circle indicating a loop. The third diagram is a loop diagram with external lines 1, 2, 3, 4 and internal lines 1, 2, 3, 4, with a blue circle indicating a loop and a blue arrow labeled k .

$$= -\lambda_0 + \frac{\lambda_0^2}{2} \int_A \frac{d^4 k}{(2\pi)^4} \left[\frac{1}{k^2 + m_0^2} \cdot \frac{1}{(k+p_1+p_2)^2 + m_0^2} \right. \\ \left. + \frac{1}{k^2 + m_0^2} \cdot \frac{1}{(k+p_1+p_4)^2 + m_0^2} + \frac{1}{k^2 + m_0^2} \cdot \frac{1}{(k+p_1+p_3)^2 + m_0^2} \right]$$

* We use the convention that all external momenta is flowing inwards

$$\langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle = \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle + \langle \phi_1 \phi_2 \phi_3 \phi_4 e^{iH_{int}} \rangle$$

$$H_{ant} = \frac{10}{46} \text{ f4 vacuum}$$

$$= \frac{1}{4!} X = \int d^4\omega (-i) \phi_1 \phi_2 \phi_3 \phi_4$$

 $+ \frac{\lambda^2}{8!} \int d^4\omega d^4y \phi_\omega \phi_\omega \phi_\omega \phi_\omega \phi_y \phi_y \phi_y \phi_y \phi_1 \phi_2 \phi_3 \phi_4$
 $= \frac{16^2}{8!} \int d^4\omega d^4y \int \frac{d^4 p_1}{(2\pi)^4} \frac{e^{-ik_0(\omega-x_1)}}{p_*^2 + m_0^2} x \dots$

$$\frac{\lambda_0^2}{2!} \int d^4\omega \int d^4\gamma \left(\prod_{i=1}^4 \int \frac{d^4 p_i}{(2\pi)^4} \right) \frac{e^{-ip_1(\omega-x_1)} e^{-ip_2(\omega-x_2)} e^{-ip_3(\gamma-x_3)} e^{-ip_4(\gamma-x_4)}}{\prod_{i=1}^4 (p_i^2 + m_0^2)}$$

$$\sqrt{\int \frac{d^4 k}{k^2} \frac{d^4 p}{p^2} \quad -i k_1 (\omega - \gamma) \quad -i k_2 (\omega - \gamma)}$$

$$= \left(\frac{1}{(2\pi)^4} \int \frac{d^4 p_i}{(2\pi)^4} \frac{1}{p_i^2 + m_0^2} \right) \left(\frac{1}{(2\pi)^4} \int \frac{d^4 q_j}{(2\pi)^4} \frac{1}{q_j^2 + m_0^2} \right) \times$$

$$\int d^4 \omega \int d^4 y \, e^{-i k_1(\omega - y)} e^{-i k_2(\omega - y)} e^{-i p_3(y - x_3)} e^{-i p_4(y - x_4)} \int d^4 y \, e^{-i y((p_3 + p_4) - (k_1 + k_2))} = (2\pi)^3 \delta^{(4)}((p_3 + p_4) - (k_1 + k_2))$$

$$\therefore (p_3 + p_4) = k_1 + k_2 = p_1 + p_2$$

$$\boxed{k_2 = (p_3 + p_4) + k_1} \leftarrow k_1 \equiv K$$

$$\frac{1}{(2\pi)^4} \int \frac{d^4 p_i}{(2\pi)^4} \left(\frac{1}{p_i^2 + m_0^2} \right) \left(e^{-i p_1 \dots} \right) \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m_0^2} \frac{1}{(k + p_1 + p_2)^2 + m_0^2}$$



$$p_1 + p_2 + k = k + p_3 + p_4$$

+ other terms as above

Since these integrals are all finite we can do a Taylor series in external momenta.

$$((k + p_1 + p_2)^2 + m_0^2)^{-1} = \frac{1}{k^2 + m_0^2} + \frac{(-1)(2)(k+p)}{((k+p)^2 + m_0^2)^2}$$

We end up w/ terms like $(\tilde{q})^2 p^2 (\tilde{q})^2$

which in real space correspond to $\sim (\partial)^2 (\partial)^2 (\partial)^2$

This means that when we start integrating out D.O.F., we will gen. all possible terms consistent w/ the symm we impose, our simplification of only considering a handful

-> All interactions are classically irrelevant which means scaling will come to the rescue and these terms will be completely swamped by the classical contribution.

~> So we do not consider them

Limit: $P_a = 0$, $a = 1, \dots, 4$:

$$\rightarrow -\lambda_0 + \frac{3\lambda_0^2}{2} \int_{\Lambda} \frac{d^4 k}{(2\pi)^4} \frac{1}{(k^2 + m_0^2)^2}$$

$$= -\lambda_0 + \lambda_0^2 \frac{3 \text{Vol}(S^3)}{2(2\pi)^4} \int_{\Lambda} d^4 k \frac{k^3}{(k^2 + m_0^2)^2}$$

$$= -\lambda_0 + \lambda_0^2 \frac{3}{32\pi^2} \left(\frac{m_0^2}{m_0^2 + \Lambda^2} - \frac{m_0^2}{m_0^2 + \Lambda^2} + \log \left(\frac{m_0^2 + \Lambda^2}{m_0^2} \right) \right)$$

$$\therefore \beta_4 = \Lambda \frac{dg_4}{d\Lambda} = \Lambda \frac{d\lambda'}{d\Lambda}$$

$$\frac{d\lambda'}{d\Lambda} = \frac{\lambda_0^2 3}{16\pi^2} \frac{m_0^2 \Lambda}{(m_0^2 + \Lambda^2)^2} + \frac{\lambda_0^2 3}{16\pi^2} \frac{m_0^2 + \Lambda^2}{(m_0^2 + \Lambda^2)} \cdot \frac{(m_0^2 + \Lambda^2) \Lambda}{(m_0^2 + \Lambda^2)^2}$$

$$= \frac{\Lambda \lambda_0^2 3}{16\pi^2} \left(\frac{m_0^2}{(m_0^2 + \Lambda^2)^2} + \frac{m_0^2 + \Lambda^2}{(m_0^2 + \Lambda^2)^2} \right)$$

$$= \frac{\lambda_0^2 3}{16\pi^2} \left[\frac{\Lambda^3}{(m_0^2 + \Lambda^2)^2} \right]$$

$$\therefore \Lambda \frac{d\lambda'}{d\Lambda} = \frac{3}{16\pi^2} \left(\frac{\Lambda^4 \lambda_0^2}{(\Lambda^2 + m_0^2)^2} \right) = \frac{3}{16\pi^2} \frac{g_4^2}{(1 + g_2)^2}$$

Recall: $\lambda_n = \Lambda^{[d_n]}$ g_n , $[d_n] = (1 - \frac{1}{2})d + n$, d^2
 where $g_4 = \lambda_0$, $g_2 = \Lambda^{-2} m_0$

• All other interactions are irrelevant so we ignore them:

* This shows that β_λ is marginally irrelevant

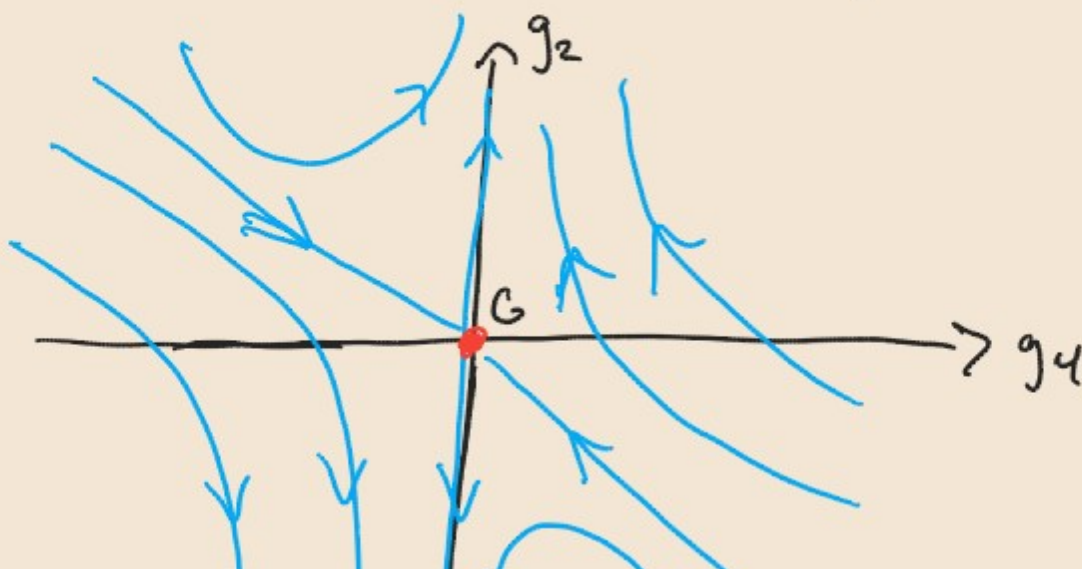
$d=4$ behaves similarly to $d>4$ in the sense

that all interactions are irrelevant and in the IR we always flow to a free theory. This is known as triviality.

The only difference between $d=4$ & $d>4$ is that the quartic interaction dies off much slower in $d=4$: it only decays logarithmically, instead of a power law.

*If we turn on the coupling we will not flow back to a Gaussian fixed pt. as there is some mixing between the coupling and the quartic coupling: i.e. the g_2 & g_4 terms both appear:

- We instead flow back to a massive theory rather than the Gaussian fixed pt.
- To get back to the Cuth fixed pt. we have to choose g_2 & g_4 appropriately!



$d < 4$:

We would like to study RG-flow for $d=1,2,3$
→ These all have relevant interactions which grow as we flow to the IR: Our weak coupling approximation is useless.

$d=1,2$: Cases are exactly solvable.

• $d=1$: We can describe the flow in terms of ordinary QM and tunneling prevents us from reaching a fixed point.

• $d=2$: We can use CFT to figure out there is actually an ∞ -# of fixed pts.

$d=3$: epsilon expansion $3 - 4 - \epsilon$

For arbitrary d :

$$\beta_2 = -2g_2 - \frac{\text{Vol}(S^{d-1})}{32\pi^4} g_4 / (1+g_2)$$

$$\beta_4 = (d-4)g_4 + \frac{3\text{Vol}(S^{d-1})}{32\pi^4} g_4^2 / (1+g_2)^2$$

• To leading order: $\text{Vol}(S^{d-1}) = 2\pi^{d/2}$

$$\beta_2 = -2g_2 - \frac{1}{16\pi^2} \frac{g_4}{1+g_2}$$

$$\beta_4 = -\epsilon g_4 + \frac{3}{16\pi^2} g_4^2 / (1+g_2)^2$$

* $\exists$ a different fixed pt to that of the Gaussian fixed pt.: The Wilson-Fisher fixed point $\beta \stackrel{!}{=} 0$:

$$-g_2^* = \frac{1}{16\pi^2} \frac{g_4^*}{1+g_2^*} = 0$$

$$-6g_4^* + \frac{3}{16\pi^2} \frac{g_2^{*2}}{(1+g_2^*)^2} = 0$$

For $g_2^* \neq 0$ & $g_4^* \neq 0$

$$-6 + \frac{3}{16\pi^2} \frac{g_4^*}{(1+g_2^*)^2} = 0$$

$$\Rightarrow \boxed{\begin{aligned} g_2^* &= -\frac{1}{6} \epsilon \\ g_4^* &= \frac{16\pi^2}{3} \epsilon \end{aligned}}$$

Since $g_4^* \sim \mathcal{O}(\epsilon)$, this fixed pt. is a weak coupling. We should now try to understand the flow about this fixed point.

To do so, we linearise the Beta f^{ns} β_n :

$$g_2 = g_2^* + \delta g_2$$

$$g_4 = g_4^* + \delta g_4$$

expand up to linear order:

$$\Delta \frac{d}{dt} \begin{pmatrix} \delta g_2 \\ \delta g_4 \end{pmatrix} = \begin{pmatrix} -2 + \frac{\epsilon}{3} & -\frac{1}{16\pi^2} (1 + \frac{\epsilon}{6}) \\ 0 & \epsilon \end{pmatrix} \begin{pmatrix} \delta g_2 \\ \delta g_4 \end{pmatrix}$$

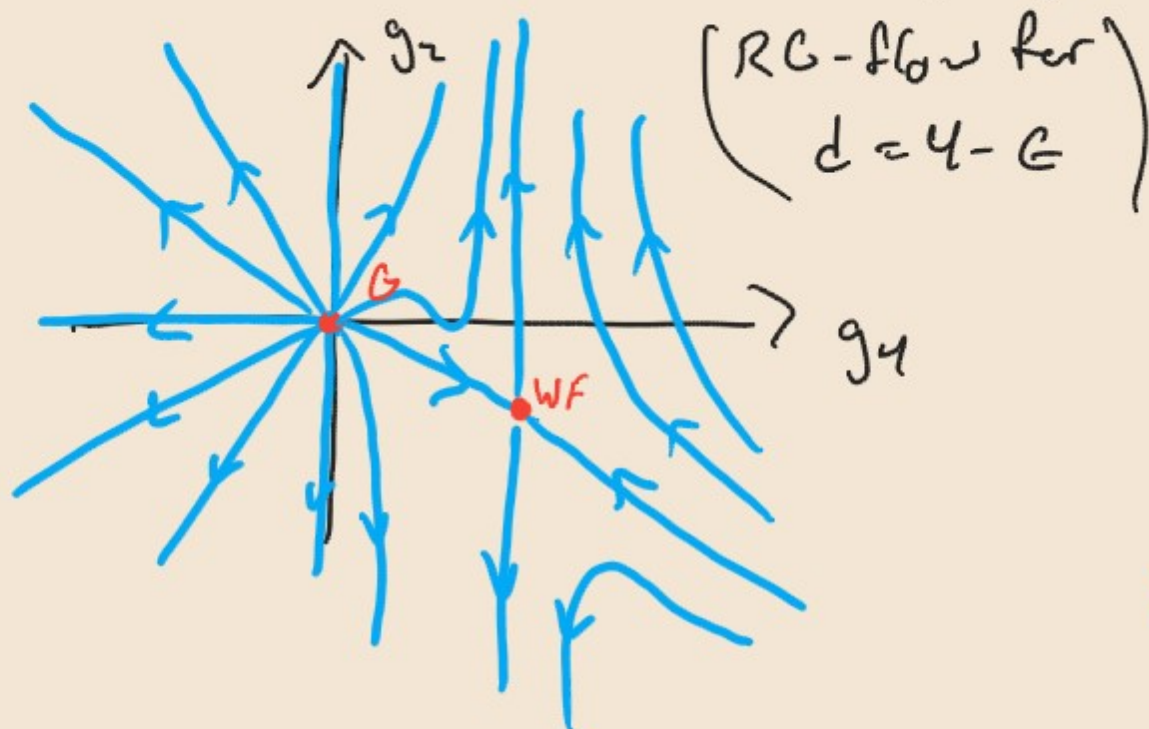
$$\bullet \beta_2 = -2(g_2^* + \delta g_2) - \left(\frac{1}{16\pi^2}\right) \frac{(g_2^* + \delta g_2)^2}{(1 + g_2^* + \delta g_2)}$$

The eigenvalues of this matrix: $\boxed{\Delta_2 = -2 + \frac{\epsilon}{3}, \Delta_4 = \epsilon}$

• Since, $\epsilon \ll 2 \Rightarrow \Delta_2 < 0$ and $\Delta_4 > 0$

so there is one relevant and one irrelevant

- The relevant direction is the mass direction
- The irr. one is some combination of g_2 & g_4



WF fixed pt: which is an interacting fixed pt.

→ This has one stable and one unstable direction

(i) Starting at the Gaussian fixed pt:
can flow in the g_2 direction

This theory is always free: i.e. we are turning coupling off

(ii) We turn on deformations that leads us to the WF fixed pt. in the IR

→ The theory becomes interacting in the IR

If we continue on we see the only way to flow from the WF is $\uparrow$ or $\downarrow$ i.e. in the g_2 direction

• The theory is free in the UV

* The theory which follows this trajectory is called asymptotically free

The above flow accurately describes the Ising model: gases in the liquid-gas phase transition.

→ Both systems exhibit second order phase transitions which characteristically are scale invariant ⇒ Correspond to fixed points

* Gaussian fixed points are unstable (IRL) we will hit a WF fixed pt. This leads to many experimental consequences.

We know that theories in the real world are interacting so we should avoid free theories such as $d=4$

QCD: Asymptotically free, meaning it can be valid at arbitrarily high energies. Also means it is more weakly coupled the higher up we go in the energy scale, and very strongly coupled in the IR

→ This leads to Confinement & hadronisation

* This a problem in knowing if the standard model has a fixed point in the UV.

Quantum Gravity:

If we take the action for GR and apply the same procedure: We find that all interactions are irrelevant. So it is hard to analyze at very high energies & the whole theory breaks down.

→ Most QG theories set a cut off before the RG flow from becoming unphysical.

- It is possible that GR actually comes from a highly non-trivial interacting fixed pt in the UV. i.e. it is asymptotically safe.

→ Def: Starting from a WF fixed pt. & turn on its relevant direction g , this theory is emerging from a fixed pt in the UV but it is an interacting fixed pt, which we call asymptotically safe.

Counterterms & The Continuum limit:

* Then: 'connect' real world to experiment.

What does this mean? We relate RG flow to an observable quantity

QFT observables:

(i) Cross-sections

(ii) Decay rates

→ Calculations of the S-matrix & LSE-reduction
i.e. $\langle \dots \rangle \rightarrow$ "observable"

* So, we now try to understand how $\langle \dots \rangle$ will flow under our previous RG prescription.

• The partition Z , which generates $\langle \dots \rangle$'s depends on

a) The source term J (trivially)

b) Couplings of the theory

c) Space time mtd $M^D \rightarrow M^D \cong \mathbb{R}^D$

d) Field Configuration space.

$\Rightarrow Z = Z(\Lambda_0, g_0, \alpha)$

If we change our scale: Nothing can happen

$$Z(\Lambda, g_n(\Lambda)) = Z(\Lambda_0, g_0, \alpha)$$

Why is this? By construction our cutoff chooses which fields we're integrating over so $Z(\Lambda, g_n(\Lambda))$ were only changing

The order in which we integrate our fields,
Starting w/ the high energy ones.
 $\Rightarrow \frac{d}{d\Lambda}(Z(\Lambda, g_\alpha(\Lambda))) = \frac{d}{d\Lambda}(Z(\Lambda_0, g_{0,\alpha}))$

But RHS is independent of Λ , so
 $\Lambda \frac{d}{d\Lambda}(Z(\Lambda, g_\alpha(\Lambda))) = 0$
 $= \Lambda \left(\frac{\partial Z}{\partial \Lambda} + \frac{\partial g_\alpha}{\partial \Lambda} \frac{\partial Z}{\partial g_\alpha} \right) = 0$
 $\therefore \left(\Lambda \frac{\partial}{\partial \Lambda} + \beta_\alpha \frac{\partial}{\partial g_\alpha} \right) Z(\Lambda, g_\alpha(\Lambda)) = 0$

$$\left(\Lambda \frac{\partial}{\partial \Lambda} + \beta_\alpha \frac{\partial}{\partial g_\alpha} \right) Z(\Lambda, g_\alpha(\Lambda)) = 0$$

$$\beta_\alpha \equiv \Lambda \left(\partial g_\alpha / \partial \Lambda \right)$$

This is known as a Callan-Symanzik (CS)
equation.

- This tells us that the coupling in the effective Action change so as to precisely balance the change in the field configuration space (cutoff).

- Use original $Z(\Lambda_0, g_{0,\alpha})$ theory:
 $\langle \hat{\phi}(x_1) \dots \hat{\phi}(x_n) \rangle = \frac{1}{Z[\phi]} \int \mathcal{D}\phi \phi(x_1) \dots \phi(x_n) e^{-S_{\Lambda_0}[\phi]}$
 $\equiv \Gamma_{\Lambda_0}^{(n)}(x_1, \dots, x_n; g_{0,\alpha})$

If $|x_i - x_j| \gg \frac{1}{\Lambda}$, then we should be able to use the low en. eff. theory.

→ Must be careful w/ field strength renorm
 $\langle \phi(x_1) \dots \phi(x_n) \rangle = \left(\int D\phi' \phi(x_1) \dots \phi(x_n) e^{-S_\Lambda[\phi']} \right) \frac{1}{Z[\phi]}$
 where $\phi' = \sqrt{Z_\phi} \phi$

$$\begin{aligned} \langle \phi(x_1) \dots \phi(x_n) \rangle &= \frac{1}{Z[\phi]} Z_\phi^{-n/2}(\Lambda) \int D\phi' \phi'(x_1) \dots \phi'(x_n) e^{-S_\Lambda[\phi']} \\ &= Z_\phi^{-n/2}(\Lambda) \Gamma_\Lambda^{(n)}(x_1, \dots, x_n; g_\Lambda(\Lambda)) \end{aligned}$$

$$\Gamma_{\Lambda_0}^{(n)}(x_1, \dots, x_n; g_{\Lambda_0}) = Z_\phi^{-n/2} \Gamma_\Lambda^{(n)}(x_1, \dots, x_n; g_\Lambda(\Lambda))$$

Infinitesimally: $\frac{d}{d\Lambda}(\quad) = \frac{d}{d\Lambda}(\quad)$

$$LHS = 0$$

$$\begin{aligned} \therefore \frac{d}{d\Lambda} \left(Z_\phi^{-n/2} \Gamma_\Lambda^{(n)}(x_1, \dots, x_n; g_\Lambda(\Lambda)) \right) &= 0 \\ &= -\frac{n}{2} Z_\phi^{-n/2-1} \left(\frac{d}{d\Lambda} Z_\phi \right) \Gamma_\Lambda^{(n)} + Z_\phi^{-n/2} \frac{d}{d\Lambda} \Gamma_\Lambda^{(n)} \\ &= -\frac{n}{2} Z_\phi^{-n/2} \frac{dZ_\phi}{d\Lambda} \Gamma_\Lambda^{(n)} + Z_\phi^{-n/2} \left(\Lambda \frac{\partial}{\partial \Lambda} + \beta_\alpha \frac{\partial}{\partial g_\alpha} \right) \Gamma_\Lambda^{(n)} = 0 \end{aligned}$$

mult. by Λ

Note: $\frac{\partial \log(Z_\phi)}{\partial \Lambda} = \frac{1}{Z_\phi} \frac{dZ_\phi}{d\Lambda}$

and so the first term is

$$Z_\phi^{-n/2} \left(-\frac{\Lambda}{2} \frac{\partial \log(Z_\phi)}{\partial \Lambda} \right) =: Z_\phi^{-n/2} n \gamma_\phi$$

γ_ϕ : Anomalous dimension

$$\therefore n \gamma_\phi \Gamma_\Lambda^{(n)} + \left(\Lambda \frac{\partial}{\partial \Lambda} + \beta_\alpha \frac{\partial}{\partial g_\alpha} \right) \Gamma_\Lambda^{(n)} = 0$$

$$\text{or } \left(\Lambda \frac{\partial}{\partial \Lambda} + \beta_\alpha \frac{\partial}{\partial g_\alpha} + n \gamma_\phi \right) \Gamma_\Lambda^{(n)}(x_1, \dots, x_n; g_\Lambda(\Lambda)) = 0$$

* At first glance, it appears that RG flow is simply a theoretical 'trick'

Is RG measurable?:

We note: M is not the mass! It is simply the coupling of the ϕ^2 term

Note RG is NOT invertable! Once we integrate out the higher energy modes, we cannot bring them back.

Counter-terms: $S_{\Lambda}(\phi) = S_B[\phi] + S_{CT}[\phi]$

where

$$S_B[\phi] := \int d^d x \left(\frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} M_0^2 \phi^2 + \frac{\lambda_0}{4!} \phi^4 \right)$$

$$S_{CT}[\phi] := \int d^d x \left(\frac{1}{2} \delta Z_\phi (\nabla \phi)^2 + \frac{1}{2} \delta M^2 \phi^2 + \frac{\delta \lambda}{4!} \phi^4 \right)$$

* Assume all the dependence of the cutoff is in the counterterm action.

→ What are we doing? We are assuming that our theory is correct at some higher energy scale. From this higher energy scale we flow back to our Λ_0 theory.

$$\Lambda' \approx \Lambda_0 + \delta \Lambda$$

so the above action is at Λ' !

* Assume Σ contribution in a diagram corresponds to a loop

- Different ways to fix the counter terms & bare couplings are called renormalisation schemes.

